

MACHINE LEARNING FOR NETWORK RELIABILITY

Machine Learning-Based Error Detection in Networks

A technical presentation on applying supervised learning models to detect, classify, and predict network anomalies in real time

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The Problem with Modern Networks

Today's networks carry massive, fast-changing traffic across distributed devices. Faults, such as congestion, packet loss, hardware failure and intrusion, surface as subtle patterns hidden in traffic data long before they cause visible outages.



Rule-based systems are static

Fixed thresholds can't adapt to evolving traffic behavior or novel fault types.



Manual monitoring doesn't scale

Human-driven diagnosis is too slow for networks generating millions of events per second.



Faults are often silent

Many errors degrade performance gradually instead of triggering an alarm.



Why It Matters

43%

of network downtime traces back to faults detected too late

<50ms

target detection latency for real-time ML pipelines

1000x

growth in monitored traffic volume over the last decade

Why Machine Learning Fits This Problem



Learns from data

Detects complex, non-linear traffic patterns that fixed rules and thresholds cannot express.



Adapts over time

Retrains on new traffic to keep pace with evolving usage patterns and emerging fault types.



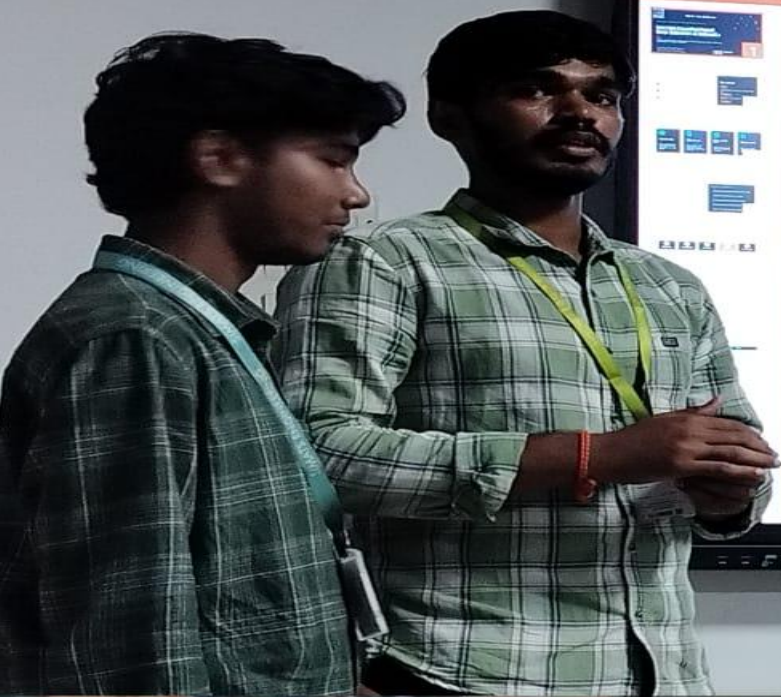
Predicts, not just reacts

Flags early warning signals before faults escalate into full outages.



Improves accuracy

Reduces false positives compared to static threshold-based alerting systems.



Team 6

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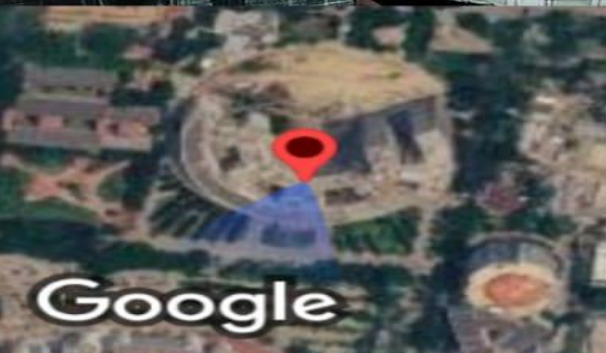
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